

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Pharm. Examination

University Theory Examination November 2016

PH219 Pharmaceutical Engineering - III

Date: 26/11/16 , Saturday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. 20
1. Which of the following is known as Noyes - Whitney equation?
[A] $dm/dt = kt$
[B] $dc/dt = k (C_s - C_b)$
[C] $dc/dt = K (a - b)$
[D] None of the above
 2. In more volatile concentrated solutions, as distance increases the concentration _____
[A] increases.
[B] decreases.
[C] remain constant.
[D] None of the above.
 3. In mass transfer, no mechanical mixing or movement is involved in _____.
[A] Conduction
[B] Convection
[C] Radiation
[D] None of the above
 4. As per BP specification, moderately fine powder pass through ____no. of sieve.
[A] 44
[B] 11
[C] 22
[D] 88
 5. When cyclone separator affects the size separation of solids suspended in a liquid such as water is called as _____
[A] liquid cyclone
[B] semi solid cyclone
[C] meta cyclone
[D] None of the above

6. By sedimentation and elutriation, _____ micrometer particle size can be determined.
 [A] 2 – 200 micrometer
 [B] 200 – 400 micrometer
 [C] 2 – 50 micrometer
 [D] 1 – 10 micrometer
7. A material is crushed in a black jaw crusher in such a way that the average size of particle is reduced from 60mm to 20mm with the consumption of energy of 17.0 Kw/(kg/s). What would be the consumption of energy needed to crush the same material of average size 75mm to an average size of 25mm. Assuming Kick's law applies?
 [A] value between 100 & 200
 [B] value between 1 & 2
 [C] value between 0.5 & 2.5
 [D] value more than 10.0
8. 5 tones = _____ lb (pound)
 [A] 11200
 [B] 1120
 [C] 11.20
 [D] 112
9. Critical speed of Ball mill is determined by following equation.
 [A] $N_c = 76.6 / \sqrt{D}$
 [B] $N_c = 76.6 + D$
 [C] $N_c = 7.6 * D$
 [D] $N_c = 76 / D$
10. One of the following is not used for size reduction.
 [A] Comminution
 [B] Grinding
 [C] Shaking
 [D] Cutting
11. The pneumatic controllers have the similar principles of _____
 [A] hydraulic controller
 [B] fluidic controller
 [C] None of the above
 [D] both
12. Squirrel cage disintegrator is also known as _____
 [A] Cage mill
 [B] Bucket mill
 [C] Bull mill
 [D] Hanging mill
13. As per IP, approximate sieve no. 10 has nominal mesh aperture size ____
 [A] 1 mm
 [B] 1.5 mm
 [C] 1.7 mm
 [D] 0.6 mm
14. 1520 mm Hg = _____ bar
 [A] 4.052
 [B] 2026
 [C] 2.026
 [D] 8.932

15. Pirani type of gauges are used for the measurement of _____
 [A] Vacuum
 [B] Volume
 [C] Weight
 [D] Level
16. Knudsen is one type of vacuum _____ gauge.
 [A] Diaphragm
 [B] Ionization
 [C] Radiometer
 [D] Compression
17. 2 atm = _____ PSI
 [A] 29.4
 [B] 34
 [C] 28
 [D] 2.8
18. 5 kg = _____ lb (pound)
 [A] 12.32
 [B] 11.025
 [C] 500
 [D] 50
19. Fire hazards due to live electrical equipments comes under class of fire
 [A] Class A
 [B] Class B
 [C] Class C
 [D] Class D
20. DDC (used in computer) means
 [A] Direct digital controller
 [B] Direct dust controller
 [C] Dust digital controller
 [D] None of the above

SECTION – II

- Q 2 Attempt any TWO of the following.**
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|----------|--|-----------|
| A | Enlist equipment used for solid-solid mixing. Explain principle, construction. | 05 |
| B | Explain principle, construction, advantages and disadvantages of Ball mill with the help of a diagram. | 05 |
| C | Describe, principle, construction, working of cyclone separator | 05 |
- Q 3 Attempt any TWO of the following.**
- | | | |
|----------|---|-----------|
| A | Write the mechanism of semisolid mixing. | 05 |
| B | Explain about Mechanical Industrial Hazards and its Safety Precautions. | 05 |
| C | Write a note on effluent treatment in pharmaceutical industry. | 05 |

SECTION – III

- Q 4 Attempt any FOUR of the following.**
- | | | |
|----------|--|-----------|
| A | Give the applications, advantages and disadvantages of planetary mixer. | 05 |
| B | How Krystal crystallizer is applied for supersaturation by evaporation? | 05 |
| C | Comment on the following :
1. The thickness of screen in hammer mill affects the product particle size. | 05 |

2. Planetary mixer can be used for mixing of semi-solids.
- D What will be the yield of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ if 500kg of the saturated solution is cooled from 110°F to 50°F . Solubility of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ per 100parts of total water at 110°F and 60°F are 37.5 and 45 parts respectively. 05
- E Comment on the following statement: Seeding is not necessary in continuous crystallizers. 05
- F What is the application of Mier's super saturation theory? 05
- Q 5 Attempt any FOUR of the following
- A Shortnote on 'the apparatus available for measuring vacuum'. 05
- B How can the theory related to the size reduction of a powder can be applied in pharma industry? 05
- C How the selection of indicating and recording instruments is done for paracetamol tablet production unit? 05
- D Classify and define the various grades of coarse and fine powders by the applications of sieves. 05
- E Differentiate Sedimentation and elutriation. 05
- F How fluid energy mill is advantageous in industry scale? 05
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